

Sheaves

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Part I

Sheaves

Chapter 1

Sheaves

1.1 (Sheaves). Let $(X, \mathcal{T})$ be a topological space. Since the topology $\mathcal{T}$ is naturally partially ordered by inclusions, we will always consider this $\mathcal{T}$ as a small subcategory of the category $\mathbf{Set}$. Let $\mathcal{C}$ be a category, set to be the category of sets if no mentioned, but we will also consider the case when $\mathcal{C}$ is the category of abelian groups or commutative unital rings.

A *presheaf* of $\mathcal{C}$ on X is just a contravariant functor $\mathcal{F} : \mathcal{T}^{\text{op}} \rightarrow \mathcal{C}$. A *sheaf* of $\mathcal{C}$ on X is a presheaf of $\mathcal{C}$ on X such that for every $U \in \mathcal{T}$ we have the equalizer in $\mathcal{C}$

$$\mathcal{F}(U) \rightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

whenever $\{U_i \rightarrow U\}$ is an open cover of U in the sense that we have the coequalizer in $\mathcal{T}$

$$\coprod_{i,j} U_i \cap U_j \rightrightarrows \coprod_i U_i \rightarrow U.$$

It is equivalent to the bijectivity of

$$\mathcal{F}(U) \rightarrow \left\{ (s_i) \in \prod_i \mathcal{F}(U_i) : s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \right\},$$

and the injectivity and the surjectivity are referred as the *locality axiom* and the *gluing axiom*. The category $\mathbf{PSh}(X)$ of presheaves of sets on X is defined to have natural transformations as morphisms, and the category $\mathbf{Sh}(X)$ of sheaves on X is defined to be the full subcategory of $\mathbf{PSh}(X)$.

For a point $x \in X$ and a presheaf $\mathcal{F}$ of a filtered cocomplete category $\mathcal{C}$ on X , the colimit $\mathcal{F}_x := \text{colim}_{U \ni x} \mathcal{F}(U)$ and its element is called a *stalk* and a *germ* at x respectively.

Let $\mathbf{LH}(X)$ be the category of local homeomorphisms over X . We construct a functor

$$\mathbf{PSh}(X) \rightarrow \mathbf{LH}(X) \rightarrow \mathbf{Sh}(X)$$

called the *sheafification*, which is left adjoint to the inclusion. The first functor to $\mathbf{LH}(X)$ is called the *étale space construction*.

- (a) There is a functor $\mathbf{PSh}(X) \rightarrow \mathbf{LH}(X) : \mathcal{F} \mapsto (\coprod_{x \in X} \mathcal{F}_x \rightarrow X)$.
- (b) There is an equivalence $\mathbf{LH}(X) \rightarrow \mathbf{Sh}(X) : Y \mapsto \Gamma(-, Y)$.

Proof. (a)

There exists a unique natural topology on $E(\mathcal{F})$ such that $p : E \rightarrow X$ is a local homeomorphism: We endow a topology on E generated by a base $\{s(U) : s \in \mathcal{F}(U), U \text{ open}\}$.

There exists a unique natural function $\mathcal{F}(U) \rightarrow \Gamma(U, E(\mathcal{F}))$ such that .. for open subsets $U \subset X$: For $x \in U$ and $s \in \mathcal{F}(U)$, we define $s_x \in p^{-1}(x)$ by the image of $\mathcal{F}(U) \rightarrow \mathcal{F}_x$ at s , and it defines a morphism of presheaves $\mathcal{F} \rightarrow \Gamma(E(\mathcal{F}))$.

(b) Let $Y \rightarrow X$ be a local homeomorphism. The functoriality of $\Gamma(-, Y)$ is clear. Let $\{U_i \rightarrow U\}$ be a cover of an open set U in X . If $s, t \in \mathcal{F}(U) = \Gamma(U, Y)$ satisfies $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$. If $\{s_i\}$ satisfies $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , then $x \mapsto s_i(x)$ is well defined independent of i with $U_i \ni x$, and is continuous.

(?) Sheafification of other target categories(I do not know the exact condition for it) can also be described as follows: since the condition for sheaves is conditioned by limits, the fully faithful inclusion $\text{Sh}(X) \rightarrow \text{PSh}(X)$ preserves limits, so it admits a left adjoint $\mathcal{F} \mapsto \mathcal{F}^\sharp$. One of the explicit construction is as follows:

$$\mathcal{F}^\sharp(U) := \text{colim}_{\{U_i \rightarrow U\} \in J(U)} \text{eq} \left(\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j) \right)$$

and $\mathcal{F}^\sharp := \mathcal{F}^{\sharp\sharp}$.

(?)

A subset $F \subset E$ defines a étale subbundle if and only if F is open in E . A covering space is nothing but a locally constant local homeomorphism. □

1.2 (Morphism of sheaves). epic and monic. The description for $\mathcal{F}(U)$ and the description for $\mathcal{F}_x$.

Let $f : X \rightarrow Y$ be a continuous map between topological spaces.

1.3 (Image functors). We have two naturally associated functors between the categories of sheaves given as follows. For a continuous map $f : X \rightarrow Y$, its *direct image functor* $f_* : \text{Sh}(X) \rightarrow \text{Sh}(Y)$ is defined by the precomposition of the opposite of $f^{-1} : \mathcal{T}_Y \rightarrow \mathcal{T}_X$, and the *inverse image functor* or the *pullback functor* $f^* : \text{Sh}(Y) \rightarrow \text{Sh}(X)$ is defined by the left adjoint to f_* .

(a) f_* is continuous, and f^* is cocontinuous.

(b) f^* is left exact in the sense that it is finitely continuous.

1.4 (Operations on sheaves). Whitney sum, constant sheaf, subsheaf
étale space descriptions

(a)

1.5 (Sheaves of rings). Every ring is assumed to be commutative and unital. A *ringed space* is a topological space X together with a sheaf $\mathcal{O}_X$ of rings on X . A *morphism* between ringed spaces is a continuous map $f : X \rightarrow Y$ together with a natural transform $f^\sharp : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ in $\text{Sh}(Y)$.

A *locally ringed space* is a ringed space X on which every stalk is a local ring. A *morphism* between locally ringed spaces X and Y is a morphism $f : X \rightarrow Y$ of ringed spaces such that the ring homomorphism $f_x^\sharp : \mathcal{O}_{Y, f(x)} \rightarrow f_{*,x}\mathcal{O}_{X,x}$ between local rings preserves the maximal ideals at each $x \in X$.

1.6 (Sheaves of module). Let X be a ringed space.

An $\mathcal{O}_X$ -module $\mathcal{F}_X$ is called *locally finite* or *finite type* if there is an open cover $\{U_i\}$ of X together with surjective ring homomorphisms $\mathcal{O}_X(U_i)^p \twoheadrightarrow \mathcal{F}_X(U_i)$ for all i . The section space $\Gamma(U, \mathcal{O}_X)$ will be also denoted by $\mathcal{O}_X(U)$.

(a) The category of sheaves of modules over $\mathcal{O}_X$ is abelian.

1.7 (Coherent sheaves). Let $(X, \mathcal{O}_X)$ be a ringed space. Consider a quasi-coherent sheaf with an exact sequence of $\mathcal{O}_X$ -modules

$$\mathcal{O}_U^q \rightarrow \mathcal{O}_U^p \rightarrow \mathcal{F}_U \rightarrow 0.$$

The *generating system* is the basis of $\mathcal{O}_X^p$, and the *relation sheaf* is the kernel of $\mathcal{O}_X^p \rightarrow \mathcal{F}_X$. We say $\mathcal{F}_X$ is *coherent* if $\mathcal{F}_X$ is of finite type and the relation sheaf is also of finite type.

(a) If $\mathcal{O}_X$ is itself a coherent module, then every locally finitely presented $\mathcal{O}_X$ -module is coherent.

1.8 (Yoga of coherent sheaves).

(a) extension principle?

(b) If a ring $\mathcal{O}$ has a split epi $\mathcal{O} \rightarrow \mathcal{O}'$ to a coherent $\mathcal{O}'$, then $\mathcal{O}$ is coherent.

Proof. Consider the following diagram in which every row is exact and K, K_1, K_2 are kernels:

$$\begin{array}{ccccccc} K & \longrightarrow & \mathcal{O}^p & \longrightarrow & \mathcal{O} & \longrightarrow & 0 \\ & & \parallel & & \downarrow \delta & & \\ K_1 & \longrightarrow & \mathcal{O}^p & \longrightarrow & \mathcal{O}' & \longrightarrow & 0 \\ & & \downarrow & & \parallel & & \\ K_2 & \longrightarrow & \mathcal{O}'^p & \longrightarrow & \mathcal{O}' & \longrightarrow & 0. \end{array}$$

Then, K_2 is finitely generated by the coherence of $\mathcal{O}'$, K_1 is finitely generated by the Schanuel lemma, and K is finitely generated by the snake lemma. $\square$

1.1

Note that affine schemes, complex model spaces, Euclidean open subsets are all locally ringed spaces.

A *scheme* is a locally ringed space such that affine open subsets form a basis.

A *complex (analytic) space* is a Hausdorff locally ringed space such that open complex model spaces form a basis...?

A *manifold* is a Hausdorff locally ringed space such that Euclidean open subsets $(U, C(U, \mathbb{R}))$ form a basis.

In the case of manifolds, a morphism of locally ringed spaces is determined by the underlying continuous map. (maybe)

They are all locally ringed. For the latter two, the residue field at every stalk is isomorphic to $\mathbb{C}$ or $\mathbb{R}$. For schemes over a field k , the relation between residue fields and k is related to the Nullstellensatz at closed points.

1.2

Chapter 2

Stacks

Chapter 3

Sites

3.1. Let $\mathcal{T}$ be a category. A *sieve* over $U \in \mathcal{T}$ is a leftward closed subcategory of the overcategory $\mathcal{T} \downarrow U$. If $\mathcal{T}$ is locally small, then it is equivalently a subfunctor of the representable functor $\text{Hom}_{\mathcal{T}}(-, U) : \mathcal{T}^{\text{op}} \rightarrow \text{Set}$.

For example, when $\mathcal{T}$ is the topology of a topological space, then since $\mathcal{T}$ is thin so that a sieve is nothing but a collection of objects of $\mathcal{T}$ satisfying some property. (a subfunctor means that $S : \mathcal{T}^{\text{op}} \rightarrow \text{Set}$ has $S(U_i) \subset \text{Hom}(U_i, U)$ and each $U_j \rightarrow U_i$ in $\mathcal{T}$ gives $S(U_i) \rightarrow S(U_j)$, if we define $\{U_i \rightarrow U\} := \bigcup_{U_i \in \mathcal{T}} S(U_i)$, then S is closed under composition in $\mathcal{T}$.)

A *Grothendieck topology* or just a *topology* on $\mathcal{T}$ is a datum that indicates which sieve is a *covering sieve*, such that

- (i) $\{U \rightarrow U\}$ is a covering sieve,
- (ii) if $\{U_i \rightarrow U\}$ is a covering sieve and if a morphism $V \rightarrow U$ admits pullbacks $V_i := U_i \times_U V$ for all U_i , then $\{V_i \rightarrow V\}$ is a covering sieve, (base change)
- (iii) if $\{U_i \rightarrow U\}$ is a covering sieve and $\{U_{ij} \rightarrow U_i\}$ is a covering sieve for each i , then $\{U_{ij} \rightarrow U\}$ is a covering sieve. (local character)

A category $\mathcal{T}$ equipped with a Grothendieck topology is called a *site*.
equivalence between topologies?

3.2. Let $\mathcal{T}$ be a site. A morphism $V \rightarrow U$ in $\widehat{\mathcal{T}}$ with $U \in \mathcal{T}$ is called a *local epimorphism* if $\{V_i \rightarrow V \rightarrow U\}$ is a covering sieve.

3.3. A *sheaf* on a site $\mathcal{T}$ is a presheaf $\mathcal{F} : \mathcal{T}^{\text{op}} \rightarrow \text{Set}$ such that for every covering sieve $S = \{U_i \rightarrow U\}$,

$$\text{Hom}(\text{Hom}(-, U), \mathcal{F}) \rightarrow \text{Hom}(S, \mathcal{F})$$

is bijective..

$$\text{Hom}(\text{Hom}(-, U), \mathcal{S}') \rightarrow \text{Hom}(S, \mathcal{S}'), \quad \forall U, S$$

is bijective?

3.1

Zariski topology

Chapter 4

Topoi